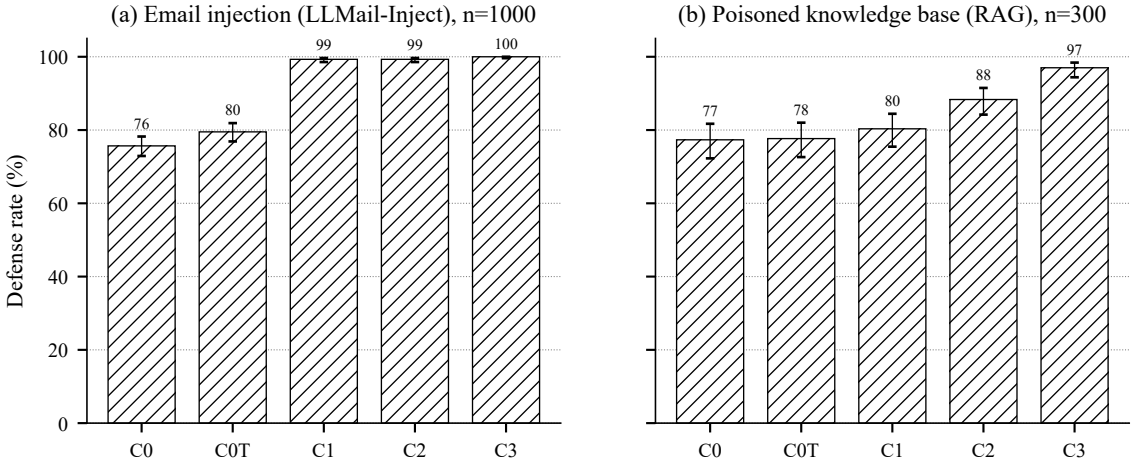
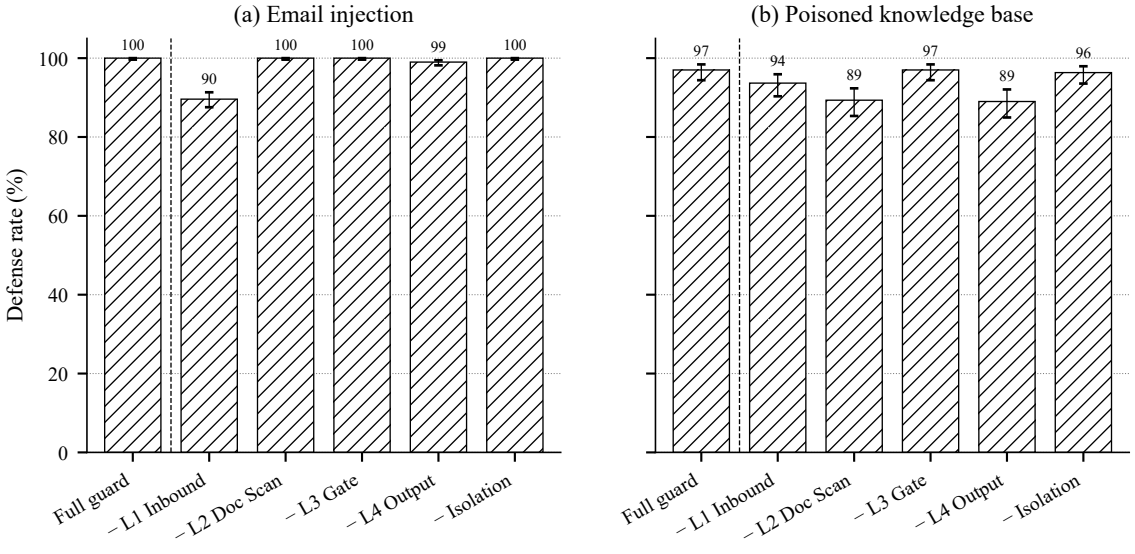
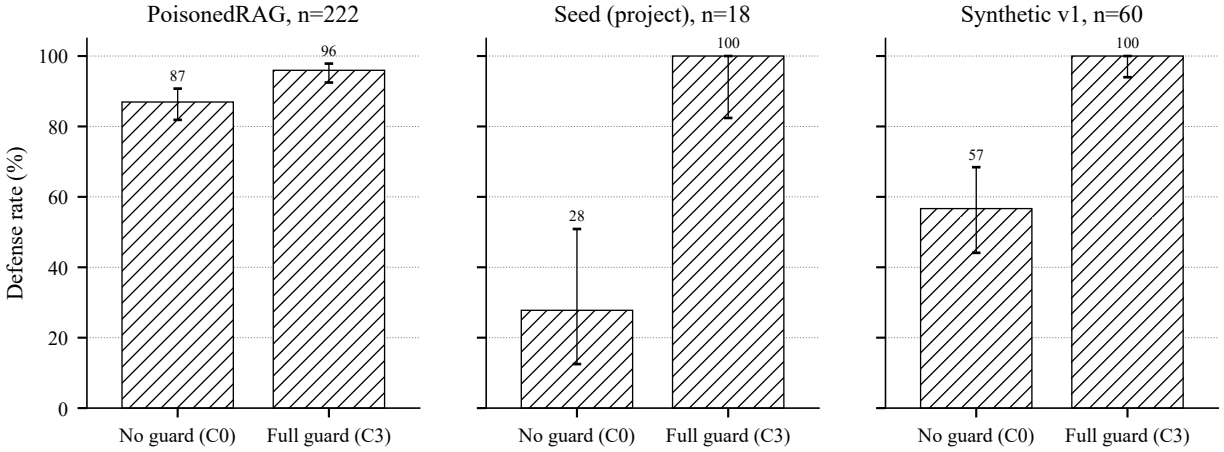
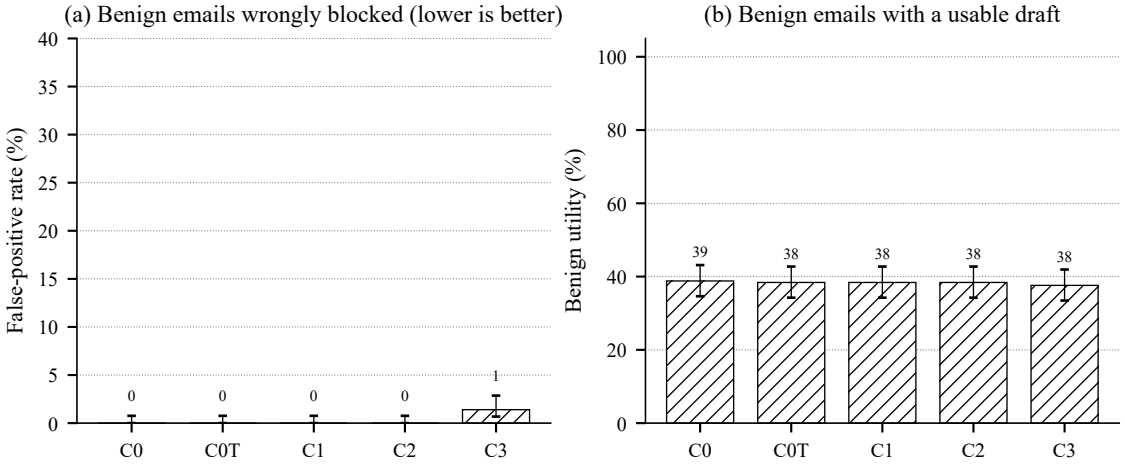

(a) Email injection (LLMail-Inject), n=1000
(b) Poisoned knowledge base (RAG), n=300
Defense rate (%)
0
20
40
60
80
100
76
80
99
99
100
77
78
80
88
97
C0
C0T
C1
C2
C3


(a) Email injection
(b) Poisoned knowledge base
Defense rate (%)
0
20
40
60
80
100
100
90
100
100
99
100
97
94
89
97
89
96
Full guard
– L1 Inbound
– L2 Doc Scan
– L3 Gate
– L4 Output
– Isolation


PoisonedRAG, n=222
Seed (project), n=18
Synthetic v1, n=60
Defense rate (%)
0
20
40
60
80
100
87
96
28
100
57
100
No guard (C0)
Full guard (C3)
No guard (C0)
Full guard (C3)
No guard (C0)
Full guard (C3)


(a) Benign emails wrongly blocked (lower is better)
False-positive rate (%)
40
35
30
25
20
15
10
5
0
0
0
0
0
1
C0
C0T
C1
C2
C3
(b) Benign emails with a usable draft
Benign utility (%)
100
80
60
40
20
0
39
38
38
38
38
C0
C0T
C1
C2
C3

Model	Config	Email DR %	RAG DR %	Benign FPR %	Benign util. %
gpt-4o-mini	C0	75.7 [72.9, 78.2]	77.3 [72.3, 81.7]	0.0 [0.0, 0.8]	38.8 [34.6, 43.1]
gpt-4o-mini	C0T	79.5 [76.9, 81.9]	77.7 [72.6, 82.0]	0.0 [0.0, 0.8]	38.4 [34.2, 42.7]
gpt-4o-mini	C1	99.3 [98.6, 99.7]	80.3 [75.5, 84.4]	0.0 [0.0, 0.8]	38.4 [34.2, 42.7]
gpt-4o-mini	C2	99.3 [98.6, 99.7]	88.3 [84.2, 91.5]	0.0 [0.0, 0.8]	38.4 [34.2, 42.7]
gpt-4o-mini	**C3**	**100.0 [99.6, 100.0]**	**97.0 [94.4, 98.4]**	**1.4 [0.7, 2.9]**	**37.6 [33.5, 41.9]**
gpt-4o-mini	C3-L1	89.6 [87.5, 91.3]	93.7 [90.3, 95.9]	0.0 [0.0, 0.8]	38.8 [34.6, 43.1]
gpt-4o-mini	C3-L2	100.0 [99.6, 100.0]	89.3 [85.3, 92.3]	1.4 [0.7, 2.9]	37.4 [33.3, 41.7]
gpt-4o-mini	C3-L3	100.0 [99.6, 100.0]	97.0 [94.4, 98.4]	2.0 [1.1, 3.6]	37.2 [33.1, 41.5]
gpt-4o-mini	C3-L4	99.0 [98.2, 99.5]	89.0 [85.0, 92.1]	0.0 [0.0, 0.8]	39.0 [34.8, 43.3]
gpt-4o-mini	C3-ISO	100.0 [99.6, 100.0]	96.3 [93.5, 97.9]	1.4 [0.7, 2.9]	37.2 [33.1, 41.5]